# VALIDATION REPORT
# OPERATIONAL QUALIFICATION
# CLINICAL MODULE

**JR Validated Environment — Clinical Module v4.0**

**Module 1: Study Design | Module 2: Diagnostic Accuracy | Module 3: Survival Analysis**

| Field | Value |
|---|---|
| Document Number | JR-VR-CLIN-001 |
| Title | Validation Report — Operational Qualification, Clinical Module |
| Validation Plan | JR-VP-CLIN-001 v4.0 (Rev 4) |
| System | JR Validated Environment — Clinical Module |
| Module Version | 4.0 |
| Document Version | 4.0 (Rev 4) |
| Execution Date | 2026-07-19 |
| OQ Result | PASS — 141 / 141 tests passed, 0 failures, 0 deviations |
| Author | Joep Rous |
| Reviewer | [Name] |
| Approver | [Name] |

## Version History

| Version | Date | Author | Description |
|---|---|---|---|
| 1.0 | 2026-07-15 | Joep Rous | Initial OQ execution report — Module 1 (Clinical Study Design). 36 / 36 test cases passed. |
| 2.0 | 2026-07-16 | Joep Rous | Scope extended to Module 2 — Diagnostic Accuracy (jrc_clinical_dx_accuracy, jrc_clinical_dx_ss, jrc_clinical_dx_roc). Re-execution of the full clinical suite: 87 / 87 test cases passed. |
| 3.0 | 2026-07-17 | Joep Rous | Scope extended to Module 3 — Survival Analysis (jrc_clinical_km, jrc_clinical_coxph). Re-execution of the full clinical suite: 123 / 123 test cases passed. |
| 4.0 | 2026-07-19 | Joep Rous | Module 2 (Diagnostic Accuracy) extended with jrc_clinical_dx_compare (paired AUC comparison, DeLong 1988). Re-execution of the full clinical suite: 141 / 141 test cases passed. |

## Approval Signatures

| Role | Name | Signature | Date |
|---|---|---|---|
| Author | Joep Rous | | 2026-07-19 |
| Reviewer | [Name] | | |
| Approver (QA) | [Name] | | |

# 1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Clinical Module, version 1.0. It provides objective evidence that all clinical study design scripts perform as specified in the OQ Validation Plan JR-VP-CLIN-001 v4.0 under the defined test conditions.

This report records the test environment, execution results for all 141 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by admin_clinical_oq.

# 2. Scope

## 2.1 In Scope

- All 3 R scripts in repos/clinical/R/ (jrc_clinical_ss_means, jrc_clinical_ss_props, jrc_clinical_ss_survival)
- Two-arm parallel-group sample sizes for superiority, non-inferiority, and equivalence (TOST) frameworks (survival: superiority and non-inferiority only)
- Numeric correctness against independently derived references (Chow, Shao & Wang 2008 ch. 3/4 normal approximations; Schoenfeld 1983 events formula)
- Framework-specific alpha and sidedness reported with the implied z-value
- Allocation ratio, dropout inflation, and sensitivity scenario tables computed in the validated R layer
- Correct rejection of invalid or incomplete designs with informative error messages
- Bypass-protection verification (RENV_PATHS_ROOT check)

## 2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- The Streamlit GUI wizard (app/clinical_design.py) — pure orchestration; all figures come from the scripts validated here
- Performance qualification (PQ) with real study protocols
- Validation of the R interpreter
- Statistical validation of the published methods themselves (Chow, Shao & Wang 2008; Schoenfeld 1983 — referenced in JR-VP-CLIN-001)

# 3. Reference Documents

| Document ID | Title | Location |
|---|---|---|
| JR-VP-CLIN-001 v4.0 | Validation Plan — OQ, Clinical Module | repos/clinical/docs/clinical_validation_plan.pdf |
| JR-VP-002 v1.0 | Validation Plan — OQ, Community Script Suite | docs/oq_validation_plan.docx |
| JR-VP-001 v1.0 | Validation Plan — Installation Qualification | docs/ |
| JR-IQ-001 | IQ Execution Evidence | docs/IQ_validation_20260311_205146.txt |
| OQ Evidence | Clinical OQ Execution Evidence (pytest output + header) | clinical_oq_execution_20260719T114744.txt |

# 4. Test Environment

| Component | Details |
|---|---|
| Execution date / time | 2026-07-19T11:47:44 |
| Host | Joeps-MacBook-Air.local |
| Operating System | [not found] |
| R Version | 4.6.0 |
| Python Version | 3.11.9 |
| pytest Version | pytest 8.3.4 |
| Test runner | repos/clinical/admin_clinical_oq |
| Test directory | repos/clinical/oq/ |
| Test data | (none — the scripts are parameter-only) |
| Execution command | bash repos/clinical/admin_clinical_oq |
| OQ evidence file | clinical_oq_execution_20260719T114744.txt |

## 5. Test Data

The clinical study design scripts are parameter-only: a sample size is computed before any study data exist, so every test case drives a script through jrrun with command-line flags alone. No CSV fixtures are used. The numeric reference values asserted by the test cases are derived independently of the scripts (documented per test case in the OQ test docstrings and in JR-VP-CLIN-001).

## 6. Test Results

All 141 test cases were executed on 2026-07-19. Results are derived from the OQ evidence file and reproduced below. The evidence file contains the complete pytest output including individual test IDs, pass/fail status, and timing.

### 6.1 Summary

| Script | Tests Executed | Result |
|---|---|---|
| jrc_clinical_ss_means | 14 | 14/14 PASS |
| jrc_clinical_ss_props | 11 | 11/11 PASS |
| jrc_clinical_ss_survival | 11 | 11/11 PASS |
| jrc_clinical_dx_accuracy | 20 | 20/20 PASS |
| jrc_clinical_dx_ss | 14 | 14/14 PASS |
| jrc_clinical_dx_roc | 17 | 17/17 PASS |
| TOTAL | 141 | 141/141 PASS |

### 6.2 Results by Test File

| Test File | TCs | Result |
|---|---|---|
| test_clinical_ss_means.py | TC-CLIN-MEANS-001..014 | 14/14 PASS |
| test_clinical_ss_props.py | TC-CLIN-PROPS-001..011 | 11/11 PASS |
| test_clinical_ss_survival.py | TC-CLIN-SURV-001..011 | 11/11 PASS |
| test_clinical_dx_accuracy.py | TC-CLIN-DXACC-001..020 | 20/20 PASS |
| test_clinical_dx_ss.py | TC-CLIN-DXSS-001..014 | 14/14 PASS |
| test_clinical_dx_roc.py | TC-CLIN-DXROC-001..017 | 17/17 PASS |
| test_clinical_km.py | TC-CLIN-KM-001..019 | 19/19 PASS |
| test_clinical_coxph.py | TC-CLIN-COX-001..017 | 17/17 PASS |
| test_clinical_dx_compare.py | TC-CLIN-DXCMP-001..018 | 18/18 PASS |

## 6.3 jrc_clinical_ss_means — 14/14 PASS

| TC ID | Description | Expected | Result |
|---|---|---|---|
| TC-CLIN-MEANS-001 | Superiority valid inputs → exit 0, report header | Exit 0, 'Clinical sample size' + 'continuous endpoint' | **PASS** |
| TC-CLIN-MEANS-002 | Superiority numeric (sd=10, δ=5, power .80, α .05/2) | n treat = 63, n control = 63, n TOTAL = 126 | **PASS** |
| TC-CLIN-MEANS-003 | Sidedness reported with z-value | '2-sided' and z = 1.9600 in output | **PASS** |
| TC-CLIN-MEANS-004 | Non-inferiority numeric (sd=8, M=4, power .90, α .025/1) | n treat = 85, n control = 85, n TOTAL = 170 | **PASS** |
| TC-CLIN-MEANS-005 | Dropout 0.15 inflation on TC-004 inputs | 'ENROLL 100 + 100 = 200' in output | **PASS** |
| TC-CLIN-MEANS-006 | Equivalence TOST numeric (sd=10, M=5, power .80) | n TOTAL = 138, 'equivalence (TOST)' in output | **PASS** |
| TC-CLIN-MEANS-007 | Allocation 2:1 on TC-002 inputs | n treat = 95, n control = 48 | **PASS** |
| TC-CLIN-MEANS-008 | Sensitivity table over SD (×0.8…1.2) | 5 scenario rows; totals 82 @ SD=8, 182 @ SD=12 | **PASS** |
| TC-CLIN-MEANS-009 | Superiority without --delta | Exit ≠ 0, 'requires --delta' | **PASS** |
| TC-CLIN-MEANS-010 | Non-inferiority without --margin | Exit ≠ 0, 'requires --margin' | **PASS** |
| TC-CLIN-MEANS-011 | Equivalence with \|δ\| ≥ margin | Exit ≠ 0, '\|delta\| < margin' | **PASS** |
| TC-CLIN-MEANS-012 | --power out of range (1.5) | Exit ≠ 0, '--power' in output | **PASS** |
| TC-CLIN-MEANS-013 | Unknown flag | Exit ≠ 0, 'Unknown argument' | **PASS** |
| TC-CLIN-MEANS-014 | Bypass protection — direct Rscript call fails | Exit ≠ 0, 'RENV_PATHS_ROOT' in output | **PASS** |

## 6.4 jrc_clinical_ss_props — 11/11 PASS

| TC ID | Description | Expected | Result |
|---|---|---|---|
| TC-CLIN-PROPS-001 | Superiority valid inputs → exit 0, report header | Exit 0, 'Clinical sample size' + 'binary endpoint' | **PASS** |
| TC-CLIN-PROPS-002 | Superiority numeric (pc .70, pt .85, power .80, α .05/2) | n treat = 118, n control = 118, n TOTAL = 236 | **PASS** |
| TC-CLIN-PROPS-003 | Non-inferiority numeric, default p-treat (pc .85, M .10) | n/arm = 201, n TOTAL = 402, 'assumed equal to control' | **PASS** |
| TC-CLIN-PROPS-004 | Equivalence TOST numeric (pc .80, M .15, power .80) | n TOTAL = 244, 'equivalence (TOST)' in output | **PASS** |
| TC-CLIN-PROPS-005 | Superiority without --p-treat | Exit ≠ 0, 'requires --p-treat' | **PASS** |
| TC-CLIN-PROPS-006 | Superiority with p-treat = p-control | Exit ≠ 0, 'different from p-control' | **PASS** |
| TC-CLIN-PROPS-007 | Non-inferiority without --margin | Exit ≠ 0, 'requires --margin' | **PASS** |
| TC-CLIN-PROPS-008 | --p-control out of range (1.2) | Exit ≠ 0, '--p-control' in output | **PASS** |
| TC-CLIN-PROPS-009 | Sensitivity table over p-control (×0.8…1.2) | 5 scenario rows (0.64…0.96); assumed row marked | **PASS** |
| TC-CLIN-PROPS-010 | Dropout 0.10 inflation on TC-003 inputs | 'ENROLL 224 + 224 = 448' in output | **PASS** |
| TC-CLIN-PROPS-011 | Bypass protection — direct Rscript call fails | Exit ≠ 0, 'RENV_PATHS_ROOT' in output | **PASS** |

## 6.5 jrc_clinical_ss_survival — 11/11 PASS

| TC ID | Description | Expected | Result |
|---|---|---|---|
| TC-CLIN-SURV-001 | Superiority valid inputs → exit 0, report header | Exit 0, 'Clinical sample size' + 'time-to-event endpoint' | **PASS** |
| TC-CLIN-SURV-002 | Superiority numeric (HR .70, power .80, α .05/2, PE .6) | Events = 247, n/arm = 206, n TOTAL = 412 | **PASS** |
| TC-CLIN-SURV-003 | Non-inferiority numeric (HR 1.0, M 1.3, α .025/1, PE .5) | Events = 457, n TOTAL = 914 | **PASS** |
| TC-CLIN-SURV-004 | --framework equivalence rejected | Exit ≠ 0, 'not offered' | **PASS** |
| TC-CLIN-SURV-005 | Superiority with HR = 1 | Exit ≠ 0, 'different from 1' | **PASS** |
| TC-CLIN-SURV-006 | NI margin ≤ 1 | Exit ≠ 0, '--margin' in output | **PASS** |
| TC-CLIN-SURV-007 | NI with HR ≥ margin | Exit ≠ 0, '--hr < --margin' | **PASS** |
| TC-CLIN-SURV-008 | Missing --event-prob | Exit ≠ 0, '--event-prob' in output | **PASS** |
| TC-CLIN-SURV-009 | Sensitivity table over P(event) (×0.8…1.2) | 5 scenario rows (0.48…0.72); assumed row marked | **PASS** |
| TC-CLIN-SURV-010 | Dropout 0.20 inflation on TC-002 inputs | 'ENROLL 258 + 258 = 516' in output | **PASS** |
| TC-CLIN-SURV-011 | Bypass protection — direct Rscript call fails | Exit ≠ 0, 'RENV_PATHS_ROOT' in output | **PASS** |

## 6.6 jrc_clinical_dx_accuracy — 20/20 PASS

| TC ID | Description | Expected | Result |
|---|---|---|---|
| TC-CLIN-DXACC-001 | Valid inputs → exit 0, report header | Exit 0, 'Diagnostic accuracy' + 'reference standard' | **PASS** |
| TC-CLIN-DXACC-002 | 2x2 contingency table from the fixture | TP 79, FP 6 / FN 16, TN 99; margins 95 / 105 / 200 | **PASS** |
| TC-CLIN-DXACC-003 | Sensitivity + Clopper-Pearson exact CI | 0.8316 (79/95), 95% CI (0.7410, 0.9006) | **PASS** |
| TC-CLIN-DXACC-004 | Specificity + Clopper-Pearson exact CI | 0.9429 (99/105), 95% CI (0.8798, 0.9787) | **PASS** |
| TC-CLIN-DXACC-005 | Overall accuracy + exact CI | 0.8900 (178/200), 95% CI (0.8382, 0.9298) | **PASS** |
| TC-CLIN-DXACC-006 | LR+ with log (Simel 1991) interval | LR+ = 14.5526, 95% CI (6.6563, 31.8163) | **PASS** |
| TC-CLIN-DXACC-007 | LR− with log (Simel 1991) interval | LR− = 0.1786, 95% CI (0.1140, 0.2799) | **PASS** |
| TC-CLIN-DXACC-008 | PPV/NPV at study prevalence, with validity caveat | PPV 0.9294, NPV 0.8609; 'study prevalence (0.4750)' + 'valid ONLY if' | **PASS** |
| TC-CLIN-DXACC-009 | --ci wilson → Wilson score interval | Estimate 0.8316 unchanged; CI (0.7438, 0.8936), inside exact | **PASS** |
| TC-CLIN-DXACC-010 | --prevalence 0.02 → Bayes-adjusted PPV/NPV | PPV (adj) 0.2290 (0.1117, 0.4637); NPV (adj) 0.9964 | **PASS** |
| TC-CLIN-DXACC-011 | --conf 0.90 narrows the interval | Estimate unchanged; both 90% bounds inside the 95% bounds | **PASS** |
| TC-CLIN-DXACC-012 | --positive resolves non-standard labels | Exit 0; sensitivity 0.8000 (4/5), specificity 0.8000 (4/5) | **PASS** |
| TC-CLIN-DXACC-013 | Rows with missing values dropped and reported | Exit 0; '4 evaluable' + '2 row(s) dropped' | **PASS** |
| TC-CLIN-DXACC-014 | Invalid --ci rejected | Exit ≠ 0, '--ci must be' | **PASS** |
| TC-CLIN-DXACC-015 | --prevalence out of range rejected | Exit ≠ 0, '--prevalence' named | **PASS** |
| TC-CLIN-DXACC- | Missing required column rejected | Exit ≠ 0, 'Missing column(s): result' | **PASS** |

| | | | |
|---|---|---|---|
| 016 | | | |
| TC-CLIN-DXACC-017 | Duplicate subject id rejected | Exit ≠ 0, 'Duplicate id(s) found' | **PASS** |
| TC-CLIN-DXACC-018 | Unrecognised labels rejected | Exit ≠ 0, 'does not recognise' | **PASS** |
| TC-CLIN-DXACC-019 | No reference-negative subjects rejected | Exit ≠ 0, 'specificity is undefined' | **PASS** |
| TC-CLIN-DXACC-020 | Bypass protection — direct Rscript call fails | Exit ≠ 0, 'RENV_PATHS_ROOT' in output | **PASS** |

## 6.7 jrc_clinical_dx_ss — 14/14 PASS

| TC ID | Description | Expected | Result |
|---|---|---|---|
| TC-CLIN-DXSS-001 | Precision valid inputs → exit 0, report header | Exit 0, 'Clinical sample size' + 'diagnostic accuracy' | **PASS** |
| TC-CLIN-DXSS-002 | Precision numeric — Buderer (1996) worked example | n ref+ = 139, n ref− = 73, N TOTAL = 1383 | **PASS** |
| TC-CLIN-DXSS-003 | Hypothesis numeric (goals .80/.90, power .80, α .025/1) | n ref+ = 108, n ref− = 239, N TOTAL = 1075 | **PASS** |
| TC-CLIN-DXSS-004 | Binding arm identified | 'Binding arm : sensitivity (reference-positive)' | **PASS** |
| TC-CLIN-DXSS-005 | Rarer prevalence raises enrolment, not arm size | n ref+ = 139 unchanged; N TOTAL = 2766 at prevalence 0.05 | **PASS** |
| TC-CLIN-DXSS-006 | Sensitivity table over prevalence (×0.5…2.0) | Exit 0; 5 rows, assumed row marked '<- assumed' | **PASS** |
| TC-CLIN-DXSS-007 | Dropout 0.10 inflation on TC-003 inputs | 'ENROLL 1195 subjects' | **PASS** |
| TC-CLIN-DXSS-008 | Sidedness default differs by method | precision 'two-sided, z = 1.9600'; hypothesis '1-sided' | **PASS** |
| TC-CLIN-DXSS-009 | Precision without --halfwidth | Exit ≠ 0, 'requires --halfwidth' | **PASS** |
| TC-CLIN-DXSS-010 | Hypothesis without --sens-goal | Exit ≠ 0, 'requires --sens-goal' | **PASS** |
| TC-CLIN-DXSS-011 | --sens-goal ≥ --sens-expected | Exit ≠ 0, '--sens-goal must be strictly less than' | **PASS** |
| TC-CLIN-DXSS-012 | --prevalence out of range (1.2) | Exit ≠ 0, '--prevalence' named | **PASS** |
| TC-CLIN-DXSS-013 | Invalid --method rejected | Exit ≠ 0, '--method must be one of' | **PASS** |
| TC-CLIN-DXSS-014 | Bypass protection — direct Rscript call fails | Exit ≠ 0, 'RENV_PATHS_ROOT' in output | **PASS** |

## 6.8 jrc_clinical_dx_roc — 17/17 PASS

| TC ID | Description | Expected | Result |
|---|---|---|---|
| TC-CLIN-DXROC-001 | Valid inputs → exit 0, report header | Exit 0, 'ROC analysis'; 80 ref+ / 120 ref− | **PASS** |
| TC-CLIN-DXROC-002 | Two-panel PNG written to ~/Downloads/ | *_jrc_clinical_dx_roc.png created during the run | **PASS** |
| TC-CLIN-DXROC-003 | PUBLISHED ANCHOR — Hanley & McNeil (1982) Table 1 | 51 ref+ / 58 ref−; AUC = 0.8932 = published 0.893 (3 dp) | **PASS** |
| TC-CLIN-DXROC-004 | AUC, DeLong SE and confidence interval | AUC 0.8268, SE 0.0294, 95% CI (0.7691, 0.8845) | **PASS** |
| TC-CLIN-DXROC-005 | Tie-aware AUC kernel (ψ = 0.5 on ties) | 12 ref+ / 14 ref−; AUC = 0.5952 | **PASS** |
| TC-CLIN-DXROC-006 | Youden-optimal operating point | J = 0.5125, sensitivity 0.5875, specificity 0.9250 | **PASS** |
| TC-CLIN-DXROC-007 | Cutoff is an observed value with an explicit rule | 'Cutoff : score >= 2.192 => positive' | **PASS** |
| TC-CLIN-DXROC- | Test of AUC = 0.5 | 'H0: AUC = 0.5' and z = 11.1017 | **PASS** |

| | | | |
|---|---|---|---|
| 008 | | | |
| TC-CLIN-DXROC-009 | --direction lower reflects AUC about 0.5 | AUC 0.1732; 0.8268 + 0.1732 = 1 within 1e-9 | **PASS** |
| TC-CLIN-DXROC-010 | Degenerate cutoff reported as none, + AUC<0.5 warning | Exit 0; 'Cutoff : none', 'AUC < 0.5', no infinite threshold printed | **PASS** |
| TC-CLIN-DXROC-011 | --ci-method logit interval | AUC unchanged 0.8268; 95% CI (0.7614, 0.8772) | **PASS** |
| TC-CLIN-DXROC-012 | --conf 0.90 narrows the interval | AUC unchanged; 90% CI (0.7784, 0.8752), inside the 95% CI | **PASS** |
| TC-CLIN-DXROC-013 | Invalid --direction rejected | Exit ≠ 0, '--direction must be' | **PASS** |
| TC-CLIN-DXROC-014 | Missing required column rejected | Exit ≠ 0, 'Missing column(s): score' | **PASS** |
| TC-CLIN-DXROC-015 | Duplicate subject id rejected | Exit ≠ 0, 'Duplicate id(s) found' | **PASS** |
| TC-CLIN-DXROC-016 | No reference-negative subjects rejected | Exit ≠ 0, 'At least 2 reference-negative' | **PASS** |
| TC-CLIN-DXROC-017 | Bypass protection — direct Rscript call fails | Exit ≠ 0, 'RENV_PATHS_ROOT' in output | **PASS** |

## 6.9 jrc_clinical_km — 19/19 PASS

| TC ID | Description | Expected | Result |
|---|---|---|---|
| TC-CLIN-KM-001 | Valid inputs → exit 0, report header | Exit 0, 'Kaplan-Meier survival analysis', 23 subjects | **PASS** |
| TC-CLIN-KM-002 | Two-panel PNG written | *_jrc_clinical_km.png created | **PASS** |
| TC-CLIN-KM-003 | PUBLISHED ANCHOR — Miller (1981) medians | Maintained median 31, non-maintained 23 | **PASS** |
| TC-CLIN-KM-004 | PUBLISHED ANCHOR — Miller (1981) log-rank | $\chi^2$ = 3.3964 (pub. 3.4), df 1, p = 0.065 (0.07) | **PASS** |
| TC-CLIN-KM-005 | Log-rank observed vs expected | Maint. 7/10.69, non-maint. 11/7.31 | **PASS** |
| TC-CLIN-KM-006 | Ungrouped overall median + CI | n 23, median 27, 95% CI (18, 45) | **PASS** |
| TC-CLIN-KM-007 | Survival at time point | S(30) = 0.6136 and 0.2917 | **PASS** |
| TC-CLIN-KM-008 | Confidence level narrows median CI | 90% CI (18, 43) inside 95% (18, 45) | **PASS** |
| TC-CLIN-KM-009 | Rho-weighted (Peto) differs from log-rank | Peto $\chi^2$ = 2.7793, ≠ log-rank 3.3964 | **PASS** |
| TC-CLIN-KM-010 | Event-label override | Exit 0, 6 subjects (km_yesno, --event-positive yes) | **PASS** |
| TC-CLIN-KM-011 | Yes/no labels auto-detected | Exit 0, 3 events | **PASS** |
| TC-CLIN-KM-012 | Median CI reported NA when not reached | Maintained median CI upper = NA | **PASS** |
| TC-CLIN-KM-013 | Missing required column rejected | Exit ≠ 0, 'Missing column(s)' | **PASS** |
| TC-CLIN-KM-014 | Duplicate id rejected | Exit ≠ 0, 'Duplicate id(s) found' | **PASS** |
| TC-CLIN-KM-015 | All-censored data rejected | Exit ≠ 0, 'No events observed' | **PASS** |
| TC-CLIN-KM-016 | Single-group comparison rejected | Exit ≠ 0, 'only one distinct value' | **PASS** |
| TC-CLIN-KM-017 | Unrecognised event labels rejected | Exit ≠ 0, 'does not recognise' | **PASS** |
| TC-CLIN-KM-018 | Non-positive time rejected | Exit ≠ 0, 'strictly positive' | **PASS** |
| TC-CLIN-KM-019 | Bypass protection — direct Rscript call fails | Exit ≠ 0, 'RENV_PATHS_ROOT' | **PASS** |

## 6.10 jrc_clinical_coxph — 17/17 PASS

| TC ID | Description | Expected | Result |
|---|---|---|---|

| | | | |
|---|---|---|---|
| TC-CLIN-COX-001 | Valid inputs → exit 0, report header | Exit 0, 'Cox proportional-hazards', 228 subj / 165 events | **PASS** |
| TC-CLIN-COX-002 | Schoenfeld PNG written | *_jrc_clinical_coxph.png created | **PASS** |
| TC-CLIN-COX-003 | PUBLISHED ANCHOR — Therneau sex HR | sex HR 0.5986, 95% CI (0.4311, 0.8311) | **PASS** |
| TC-CLIN-COX-004 | PUBLISHED ANCHOR — Therneau age HR | age HR 1.0172, 95% CI (0.9990, 1.0357) | **PASS** |
| TC-CLIN-COX-005 | Harrell's concordance | C = 0.6029 | **PASS** |
| TC-CLIN-COX-006 | Global LR and Wald tests | LR $\chi^2$ 14.1231, Wald $\chi^2$ 13.4700 | **PASS** |
| TC-CLIN-COX-007 | PH assumption OK when it holds | 'PH assumption OK', global p = 0.2502 | **PASS** |
| TC-CLIN-COX-008 | PUBLISHED ANCHOR — PH violation flagged | 'PH assumption VIOLATED', p = 0.000316 (veteran/karno) | **PASS** |
| TC-CLIN-COX-009 | Factor covariate | sex_labelmale HR 1.7007 (= 1/0.5986) | **PASS** |
| TC-CLIN-COX-010 | Missing covariate rows dropped | 227 fitted, 'Dropped rows : 1' | **PASS** |
| TC-CLIN-COX-011 | Breslow ties differ from Efron | sex HR 0.5990 (breslow) ≠ 0.5986 (efron) | **PASS** |
| TC-CLIN-COX-012 | Confidence level narrows HR CI | 90% CI (0.4545, 0.7884) inside 95% | **PASS** |
| TC-CLIN-COX-013 | Missing covariate column rejected | Exit ≠ 0, 'Missing column(s)' | **PASS** |
| TC-CLIN-COX-014 | --covariates required | Exit ≠ 0, '--covariates is required' | **PASS** |
| TC-CLIN-COX-015 | Duplicate id rejected | Exit ≠ 0, 'Duplicate id(s) found' | **PASS** |
| TC-CLIN-COX-016 | Fewer than two events rejected | Exit ≠ 0, 'Fewer than 2 events' | **PASS** |
| TC-CLIN-COX-017 | Bypass protection — direct Rscript call fails | Exit ≠ 0, 'RENV_PATHS_ROOT' | **PASS** |

## 6.11 jrc_clinical_dx_compare — 18/18 PASS

| TC ID | Description | Expected | Result |
|---|---|---|---|
| TC-CLIN-DXCMP-001 | Valid inputs → exit 0, report header | Exit 0, 'Paired ROC comparison', 70 ref+ / 100 ref- | **PASS** |
| TC-CLIN-DXCMP-002 | Two-panel PNG written | *_jrc_clinical_dx_compare.png created | **PASS** |
| TC-CLIN-DXCMP-003 | ORACLE ANCHOR — per-test AUCs (pROC) | AUC A 0.8681 (0.8138,0.9225); AUC B 0.6670 (0.5851,0.7489) | **PASS** |
| TC-CLIN-DXCMP-004 | Difference A-B with CI | +0.2011 (95% CI +0.1148, +0.2875) | **PASS** |
| TC-CLIN-DXCMP-005 | Paired DeLong test | z 4.5640, p 5.019e-06 (bit-identical to pROC roc.test) | **PASS** |
| TC-CLIN-DXCMP-006 | Between-AUC correlation | 0.2468 | **PASS** |
| TC-CLIN-DXCMP-007 | Significance statement | 'test_a has the higher AUC ... significant' | **PASS** |
| TC-CLIN-DXCMP-008 | --conf 0.90 narrows difference CI | 90% CI (+0.1287, +0.2736) | **PASS** |
| TC-CLIN-DXCMP-009 | Logit per-test CI differs, in (0,1) | AUC A logit CI (0.8037, 0.9137) | **PASS** |
| TC-CLIN-DXCMP-010 | Direction lower reflects both, flips z | AUC A 0.1319, B 0.3330; z -4.5640 | **PASS** |
| TC-CLIN-DXCMP-011 | --tests names the two columns | 'A = test_a vs B = test_b' | **PASS** |
| TC-CLIN-DXCMP-012 | Two columns auto-inferred | test_a vs test_b without --tests | **PASS** |
| TC-CLIN-DXCMP-013 | Ambiguous columns rejected | Exit ≠ 0, 'Could not infer the two test columns' | **PASS** |
| TC-CLIN-DXCMP-014 | Named column absent rejected | Exit ≠ 0, 'Test column(s) not found: nosuchcol' | **PASS** |
| TC-CLIN-DXCMP- | Missing reference rejected | Exit ≠ 0, 'Missing column(s): reference' | **PASS** |

| | | | |
|---|---|---|---|
| 015 | | | |
| TC-CLIN-DXCMP-016 | Duplicate id rejected | Exit ≠ 0, 'Duplicate id(s) found' | **PASS** |
| TC-CLIN-DXCMP-017 | No reference-negative rejected | Exit ≠ 0, 'At least 2 reference-negative' | **PASS** |
| TC-CLIN-DXCMP-018 | Bypass protection — direct Rscript call fails | Exit ≠ 0, 'RENV_PATHS_ROOT' | **PASS** |

## 7. Deviations

No deviations were identified during OQ execution. All 141 test cases passed.

## 8. Requirements Traceability Matrix

User requirements UR-CLIN-001 … UR-CLIN-141 map 1:1, in order, to test cases TC-CLIN-MEANS-001..014, TC-CLIN-PROPS-001..011, TC-CLIN-SURV-001..011, TC-CLIN-DXACC-001..020, TC-CLIN-DXSS-001..014 and TC-CLIN-DXROC-001..017 (see JR-VP-CLIN-001 Rev 2 for the full matrix). UR-CLIN-001 … UR-CLIN-036 are unchanged from Rev 1 and retain their original numbering; Module 2 appends UR-CLIN-037 … UR-CLIN-087. Grouped by capability:

| Requirement group | Capability | Test Cases |
|---|---|---|
| UR-CLIN-001..008 | jrc_clinical_ss_means — frameworks, numerics, allocation, dropout, sensitivity | TC-CLIN-MEANS-001..008 |
| UR-CLIN-009..013 | jrc_clinical_ss_means — input validation | TC-CLIN-MEANS-009..013 |
| UR-CLIN-014 | jrc_clinical_ss_means — bypass protection | TC-CLIN-MEANS-014 |
| UR-CLIN-015..018 | jrc_clinical_ss_props — frameworks and numerics | TC-CLIN-PROPS-001..004 |
| UR-CLIN-019..024 | jrc_clinical_ss_props — input validation, sensitivity, dropout | TC-CLIN-PROPS-005..010 |
| UR-CLIN-025 | jrc_clinical_ss_props — bypass protection | TC-CLIN-PROPS-011 |
| UR-CLIN-026..028 | jrc_clinical_ss_survival — frameworks and numerics (events → subjects) | TC-CLIN-SURV-001..003 |
| UR-CLIN-029..035 | jrc_clinical_ss_survival — input validation, sensitivity, dropout | TC-CLIN-SURV-004..010 |
| UR-CLIN-036 | jrc_clinical_ss_survival — bypass protection | TC-CLIN-SURV-011 |
| UR-CLIN-037..044 | jrc_clinical_dx_accuracy — 2x2, sens/spec/accuracy, LR± with intervals | TC-CLIN-DXACC-001..008 |
| UR-CLIN-045..049 | jrc_clinical_dx_accuracy — interval methods, Bayes-adjusted PPV/NPV, labels, missing data | TC-CLIN-DXACC-009..013 |
| UR-CLIN-050..055 | jrc_clinical_dx_accuracy — input validation | TC-CLIN-DXACC-014..019 |
| UR-CLIN-056 | jrc_clinical_dx_accuracy — bypass protection | TC-CLIN-DXACC-020 |
| UR-CLIN-057..064 | jrc_clinical_dx_ss — precision and hypothesis sizing, prevalence, dropout, sensitivity | TC-CLIN-DXSS-001..008 |
| UR-CLIN-065..069 | jrc_clinical_dx_ss — input validation | TC-CLIN-DXSS-009..013 |
| UR-CLIN-070 | jrc_clinical_dx_ss — bypass protection | TC-CLIN-DXSS-014 |
| UR-CLIN-071..073 | jrc_clinical_dx_roc — report, plot output, PUBLISHED AUC anchor (Hanley & McNeil 1982) | TC-CLIN-DXROC-001..003 |
| UR-CLIN-074..082 | jrc_clinical_dx_roc — AUC/DeLong interval, tie kernel, Youden cutoff, direction, degenerate cutoff, interval options | TC-CLIN-DXROC-004..012 |
| UR-CLIN-083..086 | jrc_clinical_dx_roc — input validation | TC-CLIN-DXROC-013..016 |
| UR-CLIN-087 | jrc_clinical_dx_roc — bypass protection | TC-CLIN-DXROC-017 |

## 9. OQ Conclusion

The Operational Qualification of the JR Validated Environment Clinical Study Design Module v1.0 is complete.

**Result: PASS**

All 141 automated test cases specified in JR-VP-CLIN-001 v1.0 were executed on 2026-07-19 and 141 passed. Results are derived from the OQ evidence file listed below.

- jrc_clinical_ss_means — two-arm sample size, continuous endpoint (superiority / non-inferiority / equivalence)
- jrc_clinical_ss_props — two-arm sample size, binary endpoint, risk difference (superiority / non-inferiority / equivalence)
- jrc_clinical_ss_survival — two-arm sample size, time-to-event endpoint, log-rank (superiority / non-inferiority)

*Evidence of test execution is archived at: clinical_oq_execution_20260719T114744.txt*

The Clinical Study Design module is released as version 1.0 of repos/clinical/. Any subsequent change to the scripts, wrappers, or help files within repos/clinical/ will require a change assessment and, if the change affects functionality covered by this OQ, re-execution of the affected test cases.